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Introducing the cell concept by both animal and plant cells: a historical and didactic approach

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Abstract: In France, as well as in several other countries, the concept of "cell" is introduced in school by two juxtaposed drawings, a plant cell and an animal cell. After indicating the dangers of this presentation, this paper focuses on the reasons behind the success of these two prototypes, through three complementary interpretations:

- A pedagogical one: in school it was easy to use onion skin and mouth epithelial cells, hence these both became prototypes of cells.

- A historical one: Schleiden was working on plant cells, and Schwann on animal cells; together they established common features between plant and animal cells (1838).

- A sociological one: since the emergence of the cell concept (19th century) until recently, zoology and botany were clearly separated research fields.

To-day, besides these traditional animal and plant cell prototypes, a general "egg-cell model" is used more and more to introduce and develop the cell concept.

1. The success and dangers of introducing the cell concept by two prototypes of animal and plant cells

To date, the cell concept is generally introduced by two juxtaposed drawings, a plant cell and an animal cell (figure 1 below), both in schools (scholar textbooks, documents distributed by the teacher) and outside school (museums, treatises, encyclopaedia,...).

And in this prototypical drawing, the plant cell is always juxtaposed to adjacent cells, while the animal cell is isolated.

These respective characteristics are not, in fact, specific to animal or plant cells:

- Monocellular organisms (Protists) are isolated cells, but can be either vegetable (*Chlorella* for instance) or animal.
- In pluricellular animals, most animal cells are juxtaposed to other cells. In the human body, for instance, only the blood cells, and some conjunctive cells, are isolated, but no one has the same morphology and internal structure as the animal cell drawn in the figure 1.
- Although many plant cells have cellulose walls and chloroplasts in their cytoplasm, some of them do not have these features.
- More generally, there are important differences between kinds of animal cells (neurone, muscular fibre, hepatocyte, fat cell, bone cell...) and also among plant cells (yeast, *Chlorella*, wood cell, etc.). These differences are more important than those between the drawn prototypes of animal and plant cells (figure 1).

INSERT HERE FIGURE 1

These images of the cell are strongly rooted in students' conceptions (in secondary school as well as at University). When I ask students to make a schema of a living cell, they either draw

a plant cell with a hexagonal form surrounded by adjacent cells, or else they draw an isolated animal cell, or sometimes they even draw both (Figure 2).

Some student drawings look like cells observed with an optical microscope, as in figure 1. But today, more and more drawings include ultrastructures (mitochondria, Golgi apparatus, and ergastoplasm) and even information coming from molecular biology (DNA as a double helix, ribosomes along a transfer RNA...).

By looking at school textbooks, these student' drawings can easily be explained.

- In some of them, the images are classical optical microscopy observations, as in figure 1 (e.g. among the books we consulted, in current textbooks in Algeria or Tunisia, or in French textbooks of the 1960s and 1970s).
- In most countries (e.g. consulted books for France, Germany, U.K., Lithuania, Estonia, Malta, but also other images in Algerian and Tunisian textbooks) the drawings of cells proposed to students in the textbooks are ultrastructural images, with conventional representation of each organelle (mitochondria, Golgi apparatus, etc), and also with some details pertaining to molecular biology. Most of these images are coloured, and several are tri-dimensional, but all show an isolated spherical animal cell and a non isolated polygonal plant cell. This modernisation of these traditional prototypes will be discussed in the last part of this paper.

INSERT HERE FIGURE 2

The example in figure 2 (chosen among hundreds drawings I gathered from Science University students), is typical of the growing influence of electron microscopy and molecular biology. The plant cell is polygonal, juxtaposed to other cells, and its organelles are those of images coming from electron microscopy or from molecular biology. The animal cell is not only isolated, but also without most of the membranous organelles (mitochondria, endoplasmic reticulum, Golgi apparatus, etc). In this animal cell, the captions give greater attention to the molecules responsible for proteins synthesis, inside two concentric membranes: the plasma membrane and the nucleus membrane.

More generally, many students' drawings reduce animal cells structure to two concentric circles (nuclear and plasmatic membranes), a varying number of other cell components as well. I called this kind of drawing "the fried-egg model", and have shown that it is often an obstacle to understanding the differentiation of cells (Clément 1988). For instance, I showed that some students could draw a chaplet of isolated round cells (and not differentiated neurones) to draw the cellular support of a reflex behaviour. Bachelard (1938) defined the "epistemological obstacle" as knowledge from every day life that prevents the learning of scientific knowledge. In the case of cells, the obstacle is not coming from every day life, but rather from previous teaching, so we call this a "didactic obstacle" (Clément 2001, 2003).

In other words, introducing the cell concept via two (animal and plant) cell prototypes can induce some didactic obstacles in the learning of students. They can think that:

- all plant cells have the same morphology and structure as the prototype of plant cell.
- all animal cells have the same morphology and structure as the prototype of animal cell.
- the main differences between animal and plant cells are their form (hexagonal or spherical), and the presence or absence of links with adjacent cells.

In summary, these two prototypes of cell can induce didactic obstacles to the understanding of the cell differentiation. In fact, there is less difference between an epidermal plant cell and an epidermal animal cell than between a neurone, a muscular fibre or a fat cell.

Nevertheless, these two archetypal drawings of the plant cell and the animal cell are still very frequently encountered today in most countries: in science exhibitions and reference books, as well as in educational documents, including school textbooks. In this paper I explore the reasons for the persistent success of these prototypes.

2. The pedagogical explanation

The first level of explanation is pedagogical.

School textbooks of several countries clearly explain how to observe cells with a microscope when they introduce the concept of cell. The kind of proposed observation is always the same:

- The plant cell is observed from the **onion epidermis**. For instance, in a Tunisian textbook, a two-pages section shows how to sample this epidermis, and how to observe it with different colouring. This is shown in part in Figure 3. The same images and explanations are present in a German textbook *Natura, Biologie für Gymnasien*, Band 3, Oberstufe. Stuttgart: Ernst Klett Verlag GmbH, 1995. Furthermore, I found it in the textbooks of several other countries (Algeria, France, ...). These onion cells are flat, and strongly linked to build the epidermis.
- In the same textbooks, the animal cells studied are coming from the **oral cavity epithelium** in the mouth. The procedures for collecting and then observing them under the microscope are described with the same details as for the onion cells. Even the classical Biology book (Campbell 1993) is using images of isolated cells of the oral cavity epithelium to illustrate some possible colouring in optical microscopy (p.118). These cells are flat and isolated because they are desquamating, just before dying. Shortly before their separation from the mouth epithelium, they constitute the upper part of this epithelium (figure 8, below).

INSERT HERE FIGURE 3

The first drawings of a plant cell juxtaposed with an animal cell that I found in science treatises were in "General Biology" books from the middle of the 20th century. In figure 1 for example, the two cells are not intended to synthesise all the plant or animal cells: they are precise examples of easily identifiable cells. The animal cell is from the mouth epithelium and the plant cell from the onion epidermis or from a flower petal epidermis. All the consulted biology treatises published during this period show the same kind of drawings of particular animal and plant cells.

In consequence, the prototypes of animal and plant cells are not theoretical models coming from the synthesis of observations of several different kinds of animal cells and several different kinds of plant cells. They are just the first examples of cells observed at school, presumably chosen because they are easy to observe in the classroom. Hence they have become the **typical animal and plant cells**, just as the cow is a typical Mammal.

The concept of "typicality" has been defined in cognitive psychology (Cordier 1990) to explain how very young children build conceptual categories from their own observations. The concept of cell is not built early on by children from their everyday observations, but later at school, when the first images of cells they encounter structure their conception of cell (Clément et al 2003).

The process is similar in a historical perspective. The typical cells drawn in Biology books in the 1950s' and 1960s' became references for researchers, university teachers, and then school teachers. Even today, in figures that introduce ultrastructural features¹ and also tri-dimensional cells², these more sophisticated drawings still imitate the typical prototypes of plant and animal cells. The characteristic features of onion and mouth cells may be more difficult to identify precisely, but figures always an isolated round (or spherical) animal cell, and a polygonal plant cell surrounded by others similar plant cells.

The question is why is it that these precise plant and animal cells became the two prototypes of cells? These two examples are easy to observe, but so are others (yeast cells, protozoa, etc)! I therefore examined the history of the cell concept in search for a possible explanation of this two-cells reference model.

3. The historical approach

There is a long history behind the origins of the cell concept. It spanned nearly two centuries: from the first observations of cells by Hooke (1667, with the first "microscope"), to the affirmation of Virchow "*Omnis cellula e cellula*" in 1855.

Several authors have analysed this history (among them: Florkin 1960, Canguilhem 1965, Horst 1987, Duchesneau 1987, Harris 1988). All these authors consider the meeting between Schleiden and Schwann in 1838 to be a decisive step in this development. Is this meeting decisive in the emergence of two prototypes of cells, an animal one and a plant one?

Schleiden described several microscopic plant structures, whereas Schwann made similar observations on microscopic animal structures. During their historic meeting in 1838, Schwann compared the cells drawn by Schleiden with his own drawings of cartilage cells and dorsal chord cells (figure 4). He found that the structure named "nucleus" was also present in the plant cells (called "nucleoblast" by Schleiden). They were then convinced that there was the same fundamental structure in plant and animal cells (figure 4), and that all animals and plants are composed of cells, hence the cellular theory was first formulated at that time.

It should be noted however that their first drawings of cells (Schleiden & Schwann 1838, Schwann 1839: figure 4) did not correspond to the two prototypes of animal and plant cells described above (e.g.: figure 1 & 2). Moreover, several animal cells drawn by Schwann (e.g. figure 4) are cartilage cells, which often resemble the prototypal aspect of plant cells, with juxtaposed cells and very thick cell wall. Schwann also described how all animal organs are made of cells, which are organised in different tissues. He defined 5 categories of tissues, and

¹ As in p.16 of Alberts et al, *Molecular Biology of the Cell*, N.Y. & London: Garland Publishing, Inc. second edition, 1989, or in the German Textbook *Natura, Biologie für Gymnasien*, Band 3, Oberstufe, Stuttgart: Ernst Klett Verlag GmbH, 1995

² As in the Lithuanian textbook, translated from Biology, 6/E by Sylvia S. Mader, 1998, McGraw-Hill Companies, Inc.

even described how egg cells divide into several cells. Finally, the isolated cell example of the mouth epithelium was not central in his descriptions, and not drawn at all in the images that I have consulted until now. On the other hand, we found some Schleiden's drawings of plant cells which have the same morphology as the onion epidermis cells (Schleiden 1844; figure 5), but never with one cell juxtaposed to portions of other cells; always several cells organised in a tissue (an epithelium: figure 5).

INSERT HERE FIGURES 4 & 5

The cellular biology books we consulted from the end of the 19th century (Virchow 1861, Carnoy 1884, Henneguy 1896) did not present the prototypes of plant and animal cells either, only groups of cells, tissues, and parts of organs. The concept of cell is introduced by the image of the egg of *Ascaris* and by images of plant tissues (Carnoy 1884 pages 186 & 187), or by different kinds of animal cells such as cartilage cells, liver cells, intestinal epithelium cells, salivary gland cells, etc (Henneguy 1896, pages 35 & s)

In consequence, the prototypes of animal and plant cells, which are largely present in the biology treatises and scholar textbooks in the second half of the 20th century (e.g. figure 1), appeared during the first half of the 20th century. Some more research is necessary to identify precisely when and where they were first published: in school textbooks or in biology reference books.

In summary, the meeting of a botanist and a zoologist, in 1838, played a decisive role in the emergence of cell theory, but they did not themselves introduced the two prototypes of cell. These prototypes, corresponding to precise examples of animal and plant cells, were introduced in the first half of the 20th century, with a pedagogical goal. But given the didactic shortcomings of this dual representation (see point 1 above), why has it remained so persistent even today?

4. The sociological approach

For several centuries, there was a deep-rooted division between Botany and Zoology, well before the emergence of the cell theory structured Biology. Thus it is not surprising that the cell theory originated by the meeting of a botanist (Schleiden) and a zoologist (Schwann). However, following this event, research on plants and on animals remained clearly separated in laboratories of Botany and Zoology respectively, even for the histological and cytological approaches.

In consequence, the success of this dual representation of cellular prototypes can be explained sociologically. The first books of "cell biology" to expose the concept of cell had to maintain the balance of influence between Botany and Zoology by presenting both prototypical models of cell: a plant cell and an animal cell. Just like all living beings had been categorised in only two groups: the vegetable kingdom and the animal kingdom. The discovery of cells presenting approximately the same structure among all the organisms of these two kingdoms was revolutionary, and a strong message to be taught.

In the sixties and seventies, I was teaching Histology and Cytology in the University of Lyon. I remember that there was an ongoing competition of sorts between the research laboratories

of "Animal Biology" and of "Plant Biology" to decide who has to assume the responsibility of teaching the cell concept to students.

In France, research laboratories are now structured differently, some of them working at the molecular and cellular levels, others at the organism or ecosystem levels. Nevertheless, this historic division between plant and animal biology is still perceivable. .

This division is also particularly evident in some Natural History Museums.

- In Copenhagen (Denmark), there are two separate buildings: one for the Zoological Museum and, a few kilometres away, another one for the Botanical Museum. They are just starting to discuss the possibility of a fusion.
- In Leiden (Netherlands), the permanent exhibition is focused on the evolution of living beings. The surface devoted to animals is exactly the same as the surface devoted to plants (the procaryotes are in the middle, the fossils at the first floor, and the human species at the top!). An interesting and modern temporary exhibition on biotechnology and bioethics is introduced by a presentation of the cell, using the two classical prototypes (animal and plant cells) !

In conclusion, the success of this dual representation of cell prototypes probably reflects the sociology of these research fields with an ancient but persisting division between animal and plant biologists.

This explanation can be corroborated by the progressive disappearance of this duality of prototypical cells, at the period where the success of molecular biology is restructuring the organisation of research laboratories. Molecular biology is more focused on the interior of the cell (procaryote or eucaryote cells, plant or animal cells), and less on differences between cells, or the organisation of cells in tissues. This new trend may explain the growing success of the egg-cell model that will be presented in the next paragraph.

This last remark can be related to the anthropologic approach of science education (Chevallard 1989, Clément 1998), which tries to explain the evolution of scientific knowledge in relation to social practices (here, molecular biology is linked to research and social practices in biotechnology and bio-industries).

5. The tenacity of prototypes

The modernisation of the traditional prototypes of animal and plant cells.

The archetypes of animal and plant cells arose from practical observations in optical microscopy. In several countries, textbooks illustrate these observations using onion epidermis and human oral cavity cells, to introduce images of cells that can easily be observed with an optical microscope.

Today, more and more science treatises, encyclopaedia, and also recent textbooks, now illustrate the concept of cell using ultrastructural images (like most of the images drawn by students, e.g. figure 2). However most of these books still introduce the cell concept using the classic prototypes of a plant cell adjacent to portions of other cells, and of an isolated animal cell, simply adding the ultrastructures of their organelles, and colours often in tri-dimensional (3D) representations (e.g. in figure 6, discussed below).

This illustrates that the modernisation of scientific knowledge can be compatible with the maintaining of traditional archetypes. The archetypes remain, but now integrate more recent scientific information.

INSERT HERE FIGURE 6

A Finish textbook (figure 6) provides an interesting example. Two pages introduce the concept of cell: one for the plant cell (p.22) and one for the animal cell (p.23). There are two images on each page, together covering half of each page. The first image is a histological view of a tissue, and the second is a 3D-colour enlargement of a typical plant cell (p.22) or a typical animal cell (p.23). These cells are not flat, in contrast with the two original prototypes. For instance (figure 6), the animal cell is spherical, and is supposed to be, by its location on the tissue, a cell of the base of the oral cavity epithelium. However none of these epithelium cells are spherical, or isolated. The basal cells of the epithelium are more or less cubic, each one having a very large nucleus. These basal cells divide to produce new cells which progressively move into the oral cavity of the mouth (figure 6). During this migration, they become flat, and only after that do they separate from the epithelium and finally become isolated. Thus these epithelial cells are never spherical nor can they be isolated cells while they remain in the epithelium.

In consequence, figure 11 is a mixture of prototypes:

- the pedagogical prototype is still present by the choice of a histological section of the mouth epithelium, traditionally used in teaching to introduce the cell concept;
- but here the animal cell is no longer a mouth epithelial cell, it is now a 3D cell, still isolated, but spherical, with a relatively small nucleus: like an ovocyte or an egg cell. The model that I called the "fried egg model" (Clément 1988, see above, in the introduction), is now a typical 3D egg model !

The success of the egg model:

Despite the fact that, in most of the books studied, the concept of cell is still introduced by the two traditional prototypes of plant and animal cells, cellular biology has more or less adopted a unique model for drawing cells (for the mitosis, meiosis, etc: e.g. figure 7) comprised of two concentric circles, the internal one for the nuclear membrane, the exterior one for the cell membrane. This is a logical convention, the success of which I interpret as also being due to the analogy with an egg cell.

In fact, a priori, other conventions would be possible such as squares, hexagons or triangles... In rare cases, schemas of cells are square, like in a Maltese textbook³ when introducing the concept of cell by a plant cell and an animal cell. But even in this case, the other schemas of cells are based on the conventional "egg model" (figure 7). And I never saw any schema where cells are, by convention, drawn as triangles !

The emotional dimension linked to the egg cell probably explains the success or the "egg model" for schematically representing any cell (animal or plant).

³ They use a U.K. book: Mackean, D.G., 1995, *GCSE Biology*. John Murray Ltd. UK

INSERT HERE FIGURE 7

In most of the modern books on cellular and molecular Biology, cells are now drawn as two concentric circles, like in figure 7. This is the new convention, the new prototype of the cell, easy to understand, and easy to describe in relation to a lot of cellular events (as in figure 7).

Nevertheless, this convention can induce didactic obstacles. One of them was given in the introduction (see point 1, above), concerning the difficulty of students to understand cell differentiation if they associate cell morphology exclusively to this conventional drawing.

Another didactic obstacle can also be identified in figure 7. Each cell is isolated and the captions suggest that the cell can decide to "*retain the power to divide*" or to "*become specialised*", which is not true. Cell specialisation is not controlled by the cell itself, but by its interactions with its surrounding (often other cells). This is a general problem in Biology Education, when the processes of interaction are not taught, but rather replaced by a single determinism, for example when DNA is the unique explanation, like God or spirit in biblical explanations (Forissier & Clément 2003). In figure 7, the determinist conception is suggested more strongly than the interaction / emergence conception. The egg-model of the cell can be associated to this determinist ideology, because it is often supposed that all the information of the adult human is contained in the egg cell.

Last remark : in some cases, the general image of the animal cell is a more or less spherical, isolated cell (as usual), but with a flagellum (Campbell 1995, p.124) - like spermatozoa ? Other animal cells with a flagellum are very rare, whereas unicellular vegetable organisms with a flagellum are more common! But I never found a prototype drawing of plant cell with a flagellum...

6. Summary and discussion

Making prototypical representation of a category of objects that vary greatly in form is a classic problem in cognitive psychology (Cordier & Denhière 1990, Cordier 2001). Several schools of thought have debated the ways in which very young children progressively establish concepts that correspond to domains.

For certain authors, children progressively synthesise the attributes of individual objects, identifying common aspects that are attributed to domains (for example characteristics of animals or of trees). For others ("theory-theory": Gopnik & Wellman 1998) the attributes of a domain are established early on, and only then the children progressively integrate in this domain singular objects possessing both these general attributes and more specific ones. We have shown that this is the case of the concept "bone" for example (Savy & Clément 2003).

Here, students construct the concept "cell" as part of their learning in school. Hence their individual construction of the concept is highly dependent on how it is introduced in teaching syllabi and in textbooks. We have seen that the cell concept has long been and continues to be introduced by a juxtaposition of two prototypical schemas of a plant cell and of an animal cell. Three complementary explanations have been given here to explain this dual representation:

1. The cell concept originated in 1839 when the botanist Schleiden and the zoologist Schwann discovered that, using microscopes, they had both observed the same structures in plant and

animal cells. However, their initial schemas of these observations do not resemble to the two prototypes that have since been adopted as a reference.

2. These two prototypes were introduced in scientific treatises and textbooks because of the ease of observing these two kinds of cells in laboratory exercises, respectively cells of the onion epidermis and cells of the human mouth epithelium. Yet today, many textbooks in many countries still refer to these two examples to introduce the cell concept, juxtaposing them as prototypical models of cell.

3. When considering the great diversity among plant cells on one hand and among animal cells on the other, as well as the essentially common aspects of the structure and functioning of these two groups of cells, it is clear that there is little scientific justification for the two prototypical models of plant and animal cells. In spite of this lack of justification, this dual prototype has remained the main reference for introducing the cell concept in teaching, internationally. Today, this model has been modernised using images from electronic microscopy, and 3D, colour representations. Two complementary explanations were proposed to explain this. Psychologically, teachers reproduce the models they themselves learned, while improving on them. Sociologically, the separation between botany and zoology has favoured the persistence of this dual approach to the same concept.

4. The recent focus on cellular and molecular biology is progressively erasing this sociological opposition between botanists and zoologist. Today the main distinction is between research centres in cellular or molecular biology on one hand, and centres focusing on organisms and ecosystems on the other, whereas both levels work indifferently on animals or plants (as well as on prokaryotes). A new prototype of cell schema has emerged today, comprising two concentric circles. Initially derived from the example of a cell from the human mouth epithelium (an isolated, dying, flat cell, that I have referred to the "egg-fried model", Clément 1988, 1998), the new prototype has evolved from the initial reference and is now the "egg-model" (figures 6 & 7). This schema is now typically represented in 3D with all its internal structures (figure 6), or in 2D as a simple framework for presenting cell processes (protein synthesis, cell division, ...: figure 7).

7. Perspectives: how to induce changes ?

The modernisation of a traditional inadequate duality of prototypes

Finally, our research presents two main trends in the way the cell concept is introduced and taught at school.

The first trend is the great modernisation of the contents being taught. Optical microscopic features are progressively replaced by ultrastructures, and by functions discovered in molecular biology. The present work did not focus on these changes, they were simply mentioned. The scientific knowledge on cellular and molecular biology is constantly renewed and reinforced by ongoing research, so it is logical that the teaching of this subject in the Universities and Schools also evolves regularly.

In contrast, the second trend, on which this paper is centred, is more conservative, showing stability throughout these other ongoing changes. The duality between plant cells and animal cells when introducing the cell concept is in effect quite stable throughout most of the period studied. These prototypic schemas had been modernised (electron microscopy, 3D, colour...),

but always maintain the duality of these prototypes, with just an adjunction of a prokaryote cell as a third prototype.

This duality of prototypic schemas is not scientifically justified (see introduction). I proposed several hypotheses to explain why these two prototypic schemas have been maintained, based on historical, pedagogical, sociological and psychological arguments respectively.

In the end, we have shown that, with the rise of a molecular approach to cellular biology, a more theoretical schema (that I call the egg model) is now generally used to represent the cell, even when the introduction of the concept of cell is still done with the dual animal and plant cell prototypes.

What are the didactic challenges of this historical / epistemological approach?

The main didactic problem is the way in which a prototype linked to a concept is introduced. If there is no explanation given regarding the origins of the prototype(s), for example for the cell, the students may be stuck with this unique reference, unable to conceive others. As explained above in the introduction, this could be a didactic obstacle to their comprehension of cellular differentiation.

Today, with the success of molecular biology in research and its influence on school textbooks, there is a real danger that this didactic obstacle be further amplified, as teaching is now more centred on intracellular processes and less on cellular differentiation and the organisation of cells in tissues and organs. To avoid this, we recommend the following measures:

1. The historical approach could be included when introducing the cell concept. This is done well in some countries, where the first drawings of cells by Hooke, as well as those of Schleiden and Schwann, are presented first, sometimes with photos of Schleiden and of Schwann (German and Tunisian school textbooks). In this sense, the concept of cell is immediately thought as a common structure of different kind of cells, at least from the first historical drawings of animal and plant cells.
2. The common features of all the cells can also be introduced by an initial presentation of different kinds of animal cells and plant cells: muscular cells, neurones, fat cells, epithelial cells, as well as animal and vegetable unicellular organisms, and different kinds of plant cells from pluricellular plant organisms.

This approach has started to be developed in the textbooks of some countries. For instance in France where, unhappily, the historical approach of the concept of cell is still absent...

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Schwann Th.: 1939, *Mikroskopische Untersuchungen*. Quoted by C. de Duve, 1987, and by V.Host 1987. A French translation of the cell theory of Schwann has been done by M.Lereboullet, and published in 1942 by Pr. Müller: 'Recherches microscopiques sur la conformité de structure et d'accroissement des animaux et des plantes. *Annales des Sciences Naturelles, Partie Zoologie*, Paris.

Virchow R.: 1861, *La pathologie cellulaire*. J.B.Baillière & Fils, Paris (2nd ed. translated from German by P.Picard)

LEGENDS OF FIGURES

Figure 1 - Two typical cells that introduce the cell concept in a French Science book
(redrawn from Oubr , Campan & Chanton, 1960, « *Biologie cellulaire* », Paris : G.Douin ed.)

Figure 2 - Drawing of a Biology student (2nd year in the University Lyon 1) to answer to the professor's question: "*Draw a living cell, with all that is important for its life*"

Figure 3 - The onion epithelial cells. From a Tunisian school textbook: Abroug et al, *Sciences Naturelles pour 3^{ me} ann e sciences exp rimentales*, Tunis: CNP (Centre National P dagogique), Minist re de l'Education, 2001

Figure 4 - Schwann's original drawing (1839): 1, 2 & 3 are plant cells, from Schleiden; the others are animal cells from Schwann; most of them are cartilage cells.

Figure 5 – Schleiden's original drawing of plant cells (from Schleiden 1944)

Figure 6 – On the left, a transversal section of the mouth epithelium, observed with optical microscope. On the right, an animal cell seeming to be one of the cells of this epithelium (with ultrastructures of organelles, not redrawn here). In fact, this animal cell is a theoretical model, the "egg-cell" model.

Totally redrawn (also with new legends) from a Finnish textbook (Saarivuori et al, 2003, *El m  ja sen monimuotoisuus*. Vsoy, Luonto, p.23)

Figure 7 - The "egg- model". Redrawn from an English textbook used in Malta (Mackean, D.G., 1995, *GCSE Biology*. John Murray Ltd. UK.). Each cell is isolated. There is no mention of interaction between cells that could explain the differentiation (and specialisation) of cells.

Figure 1

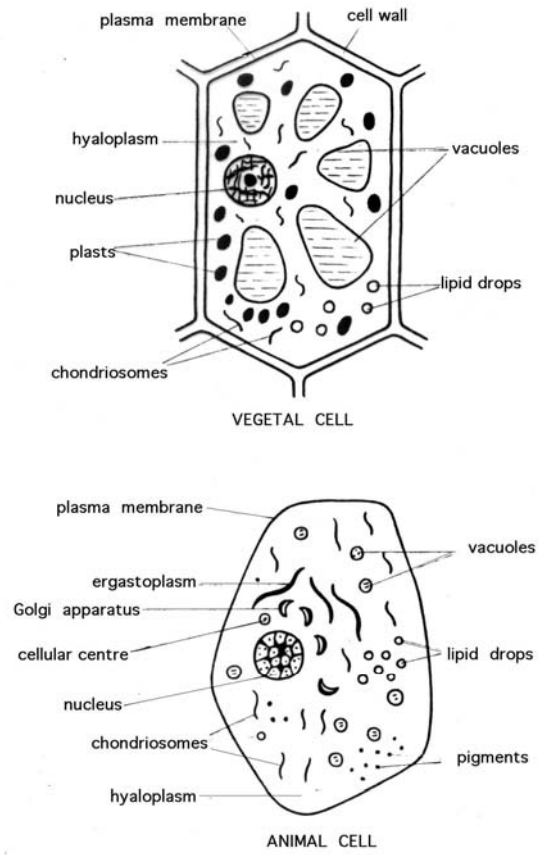


Figure 2

Figure 3

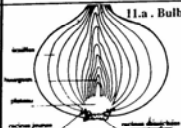
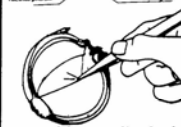
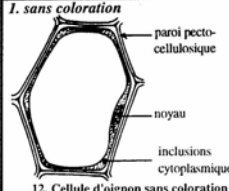
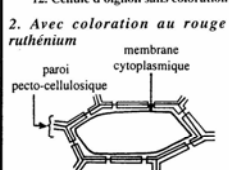
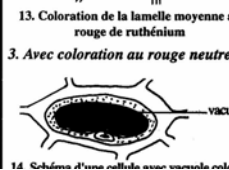
3. Etude de la cellule végétale			
Matériel utilisé	Méthodes et techniques d'étude	Aspect des cellules observées au microscope	Structures cellulaires révélées
Cellules d'oignon	11.a. Bulbe d'oignon  11.b. Prélèvement d'un lambeau d'épiderme interne d'oignon  • Couper un bulbe d'oignon. Séparer une écaille charnue. Prélever un lambeau d'épiderme interne. Couper un petit carré d'épiderme et le placer dans une goutte d'eau ou de colorant (rouge de ruthénium, rouge neutre...) entre lame et lamelle. Coloration des vacuoles. • Plongez le petit lambeau pendant 5 mn dans une solution de rouge neutre à 0,3%, rincez-le dans de l'eau distillée, montez-le entre lame et lamelle dans une goutte d'eau.	1. sans coloration  12. Cellule d'oignon sans coloration 2. Avec coloration au rouge de ruthénium  13. Coloration de la lamelle moyenne au rouge de ruthénium 3. Avec coloration au rouge neutre  14. Schéma d'une cellule avec vacuole colorée	On observe des cellules polyédriques, limitées par une membrane formée de deux couches celluloses : c'est la membrane cellulosique. • La coloration au rouge de ruthénium, permet de mettre en évidence une autre membrane de nature pectique. Elle est située entre les deux couches celluloses : c'est la membrane moyenne ou lamelle moyenne. • L'ensemble membrane cellulosique et membrane moyenne constitue la paroi pecto-cellulosique. • La coloration des cellules au rouge neutre permet de mettre en évidence des cavités remplies de liquide : ce

Figure 4

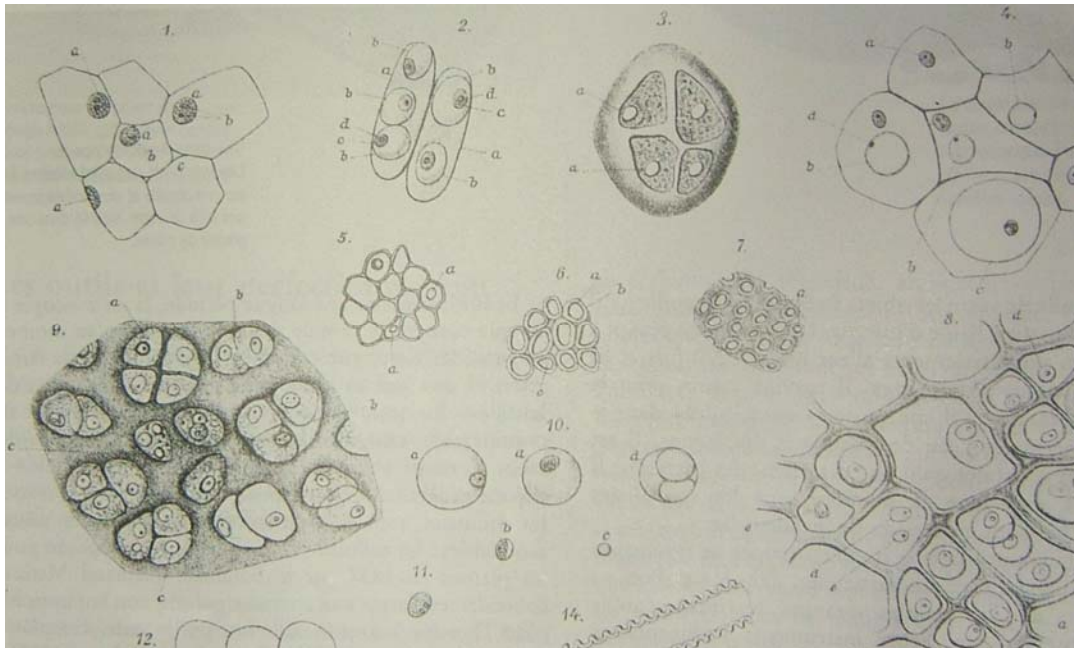


Figure 5

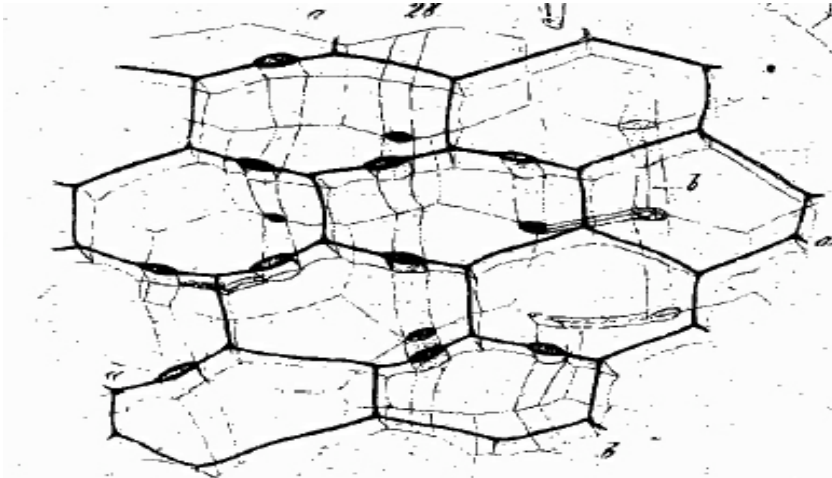


Figure 6

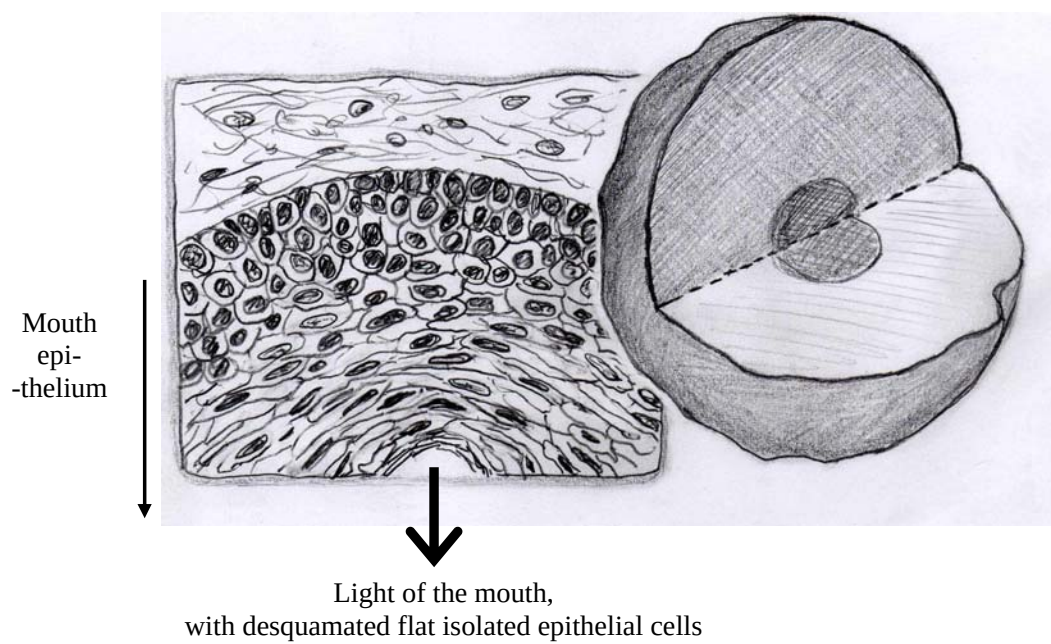


Figure 7

